Toronto Killed or Seriously Injured Collisions

What the data says once you know what a row is | Niloufar Rahmani | Source: City of Toronto Open Data, retrieved 2026-11-02 | 20,882 person records across 7,673 collisions

1. What one row is

> The file has one row per person involved in a collision, not one row per collision. The average collision involves 2.7 people. Counting rows therefore overstates every collision-level figure by roughly that factor, and the error is invisible — the numbers look plausible.

This was not assumed from the column names. Every column was tested: for each one, does its value ever differ between two people in the same collision? 30 columns never vary within a collision, which makes them facts about the collision copied onto each person row. 16 columns do vary, which makes them facts about the person. Four are identifiers.

The consequence is not academic. `acclass`, the severity of the collision, is one of the collision-level columns. Filtering rows on it and counting them answers neither of the two questions a reader would be asking.

question	naive_method	naive_answer	correct_method	correct_answer	overstated_by
How many KSI collisions are in this file?	count of rows	20882	count of distinct collision_id	7673	2.72
How many collisions were fatal?	count of rows where acclass = 'Fatal Injury'	2950	distinct collision_id where acclass = 'Fatal Injury'	1073	2.75
How many people were killed?	count of rows where acclass = 'Fatal Injury'	2950	count of rows where injury = 'Fatal'	1109	2.66
How many people were seriously injured?	count of rows where acclass = 'Non-Fatal Injury'	17911	count of rows where injury = 'Major'	7171	2.50
How many collisions involving a pedestrian?	count of rows where pedestrian is true	8511	distinct collision_id where pedestrian is true	3422	2.49
How many collisions involving a cyclist?	count of rows where cyclist is true	2185	distinct collision_id where cyclist is true	935	2.34
How many collisions flagged aggressive driving?	count of rows where aggressive is true	9319	distinct collision_id where aggressive is true	3189	2.92
How many collisions flagged distracted driving?	count of rows where distracted is true	6241	distinct collision_id where distracted is true	2248	2.78
How many collisions flagged red-light running?	count of rows where red_light is true	1700	distinct collision_id where red_light is true	476	3.57
How many collisions involving a heavy truck?	count of rows where heavy_truck is true	2813	distinct collision_id where heavy_truck is true	939	3.00
How many collisions flagged school child?	count of rows where school_child is true	4100	distinct collision_id where school_child is true	1102	3.72
How many collisions flagged older adult?	count of rows where older_adult is true	6059	distinct collision_id where older_adult is true	2188	2.77

The two correct answers in the third row are different from each other and both are right: 1,073 collisions were fatal, and 1,109 people died in them. The gap is the collisions that killed more than one person, plus 8 records discussed in section 2. Which number belongs in a headline depends on whether the sentence is about crashes or about people, and that is a decision to make deliberately rather than by accident of query.

The same trap sits under the word pedestrian. Collisions involving a pedestrian: 3,422. People present in those collisions: 8,511. Pedestrians actually involved: 3,682. Pedestrians killed: 606. All four are legitimate answers to questions that sound identical when spoken aloud.

2. Data quality

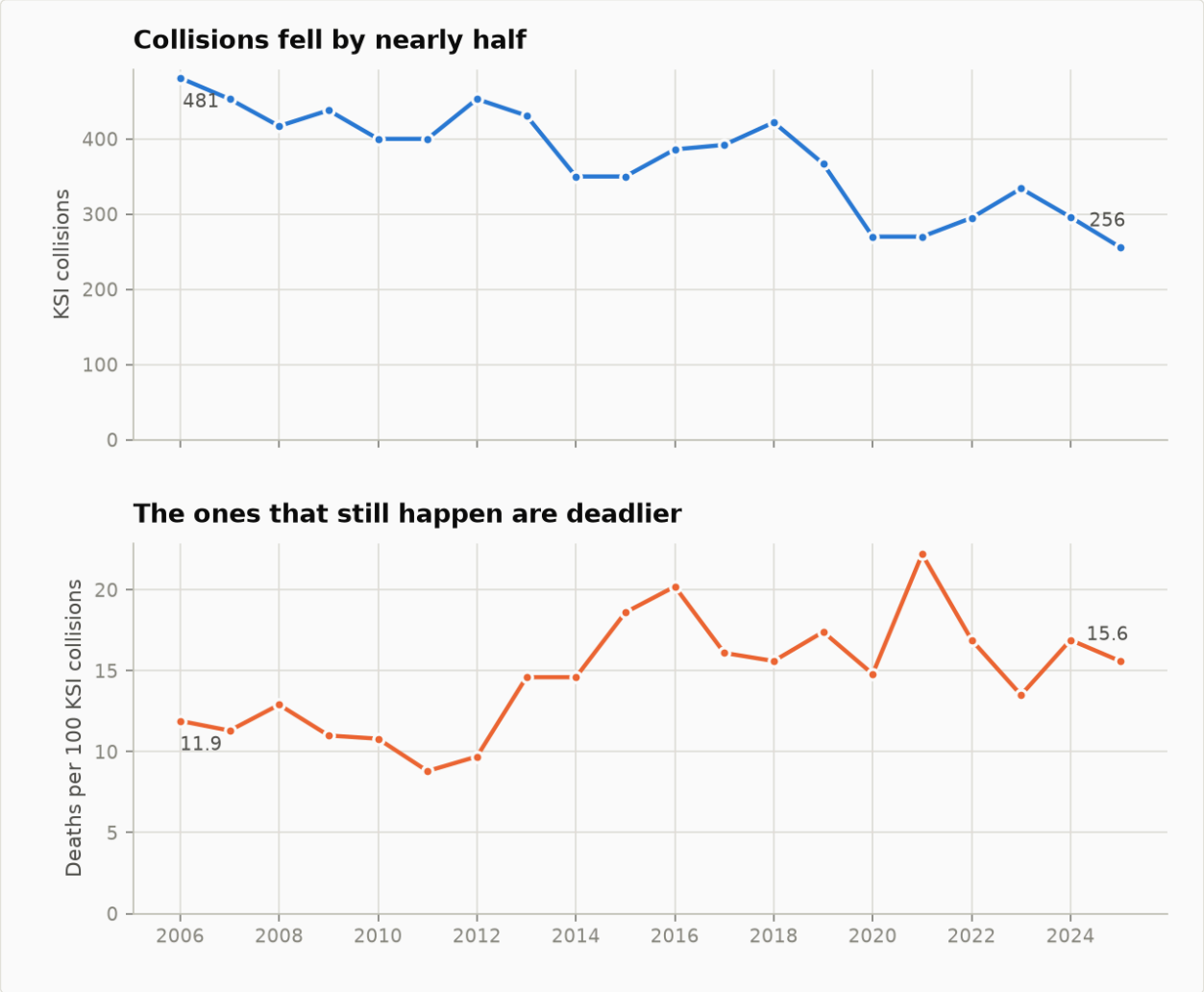
Rules were run against the extract exactly as published. The dataset is in good condition; the failures below are small, and they are recorded because a consumer needs to know about them before quoting a number, not as a criticism of the publisher.

dimension	rule	records_failed	share_failed	outcome	impact
Uniqueness	one row per (collision, vehicle, person)	0	0.00000	Pass	Confirms the stated grain. Nothing downstream is safe without it.
Completeness	acclass is populated	1	0.00005	Fail	A row with no severity class cannot be counted as fatal or non-fatal.
Completeness	coordinates are populated	3	0.00014	Fail	Rows without coordinates drop silently out of any map or spatial join.
Completeness	neighbourhood is populated	151	0.00723	Fail	Affects the neighbourhood ranking, which is built on this column.
Validity	coordinates fall inside Toronto	0	0.00000	Pass	No coordinate lands outside the city, so the geography can be trusted.
Validity	acclass uses a known severity code	2	0.00010	Fail	An unrecognised code would fall through every severity filter.
Validity	injury uses a known severity code	0	0.00000	Pass	Same risk on the person-level severity column.
Consistency	a fatal injury sits in a collision classed fatal	8	0.00038	Fail	The two severity columns disagree, so the death count and the fatal collision count cannot both be right on these rows.
Consistency	a collision classed fatal records at least one death	0	0.00000	Pass	Confirms the fatal classification is supported by a person record.
Scope	every record meets the killed-or-seriously-injured criterion	18	0.00086	Fail	Records classed property damage only do not belong in a KSI extract and will inflate any unfiltered collision count.

- 8 person records are marked injury = 'Fatal' while their collision is not classed 'Fatal Injury'. This is why the death count (1 per person record) and the fatal collision count do not reconcile to each other exactly, and the difference is precisely these rows.
- 18 person records across 8 collisions are classed 'Property Damage Only' in an extract published as killed-or-seriously-injured. They are kept in the figures here and flagged rather than dropped, because removing records from a public dataset without saying so is how two analysts end up with different totals.
- injury is missing on 47% of rows, which is expected: it is populated only where there was an injury to record. The consequence is that severity has to be counted with an equality test, never with count() or a row count.

3. Twenty years of KSI collisions

2006 to 2025. The final year in the extract is partial and is excluded from every trend in this report rather than plotted as a collapse.



KSI collisions per year, and deaths per 100 KSI collisions.

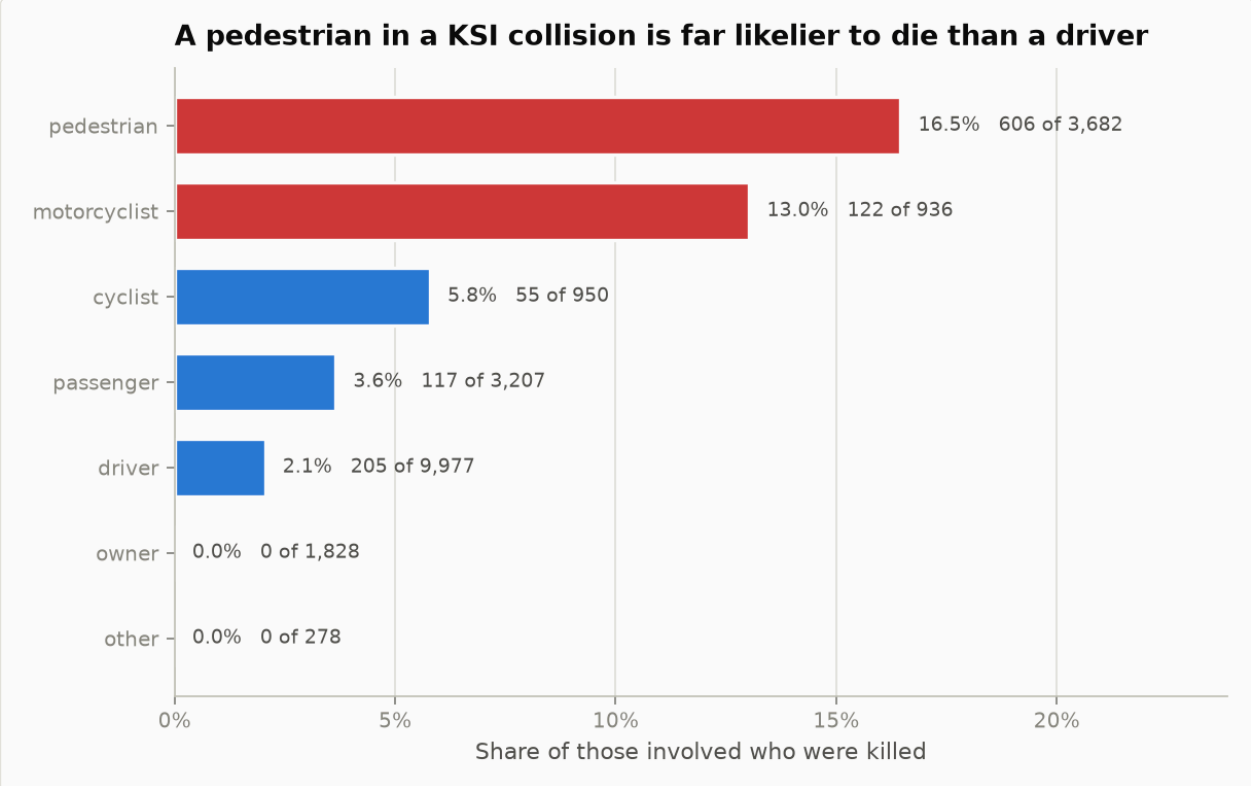
> KSI collisions fell 47% between 2006 and 2025, from 481 to 256. Deaths per 100 KSI collisions moved the other way, from 11.9 to 15.6. Fewer collisions reach this threshold than twenty years ago, and a larger share of the ones that do are fatal.

Two readings fit that pattern and this data cannot separate them. Either the collisions being prevented are disproportionately the survivable ones, or the reporting threshold for a serious injury has tightened over twenty years so that fewer non-fatal cases enter the file. A fatality is hard to misclassify; a serious injury is a judgement. Anyone using this trend to argue that roads have become more lethal per collision needs to rule the second explanation out first, and nothing in this extract does that.

year	ksi_collisions	fatal_collisions	people_involved	people_killed	people_seriously_injured	deaths_per_100_collisions
2006	481	57	1483	57	488	11.9
2007	453	47	1474	51	442	11.3
2008	417	51	1239	54	394	12.9
2009	438	45	1242	48	431	11.0
2010	400	42	1190	43	402	10.8
2011	400	35	1181	35	392	8.8
2012	453	44	1348	44	430	9.7
2013	431	63	1232	63	401	14.6
2014	350	51	916	51	330	14.6
2015	350	65	929	65	320	18.6
2016	386	76	1006	78	337	20.2
2017	392	62	980	63	352	16.1
2018	422	66	1074	66	382	15.6
2019	367	63	940	64	330	17.4
2020	270	40	640	40	246	14.8
2021	270	58	656	60	241	22.2
2022	295	48	730	50	264	16.9
2023	334	43	770	45	301	13.5
2024	296	46	725	50	266	16.9
2025	256	34	622	40	234	15.6

4. Who is being killed

This is a person-level question, so it is answered from the person rows. The rate below is the share of people involved, in that role, who died.



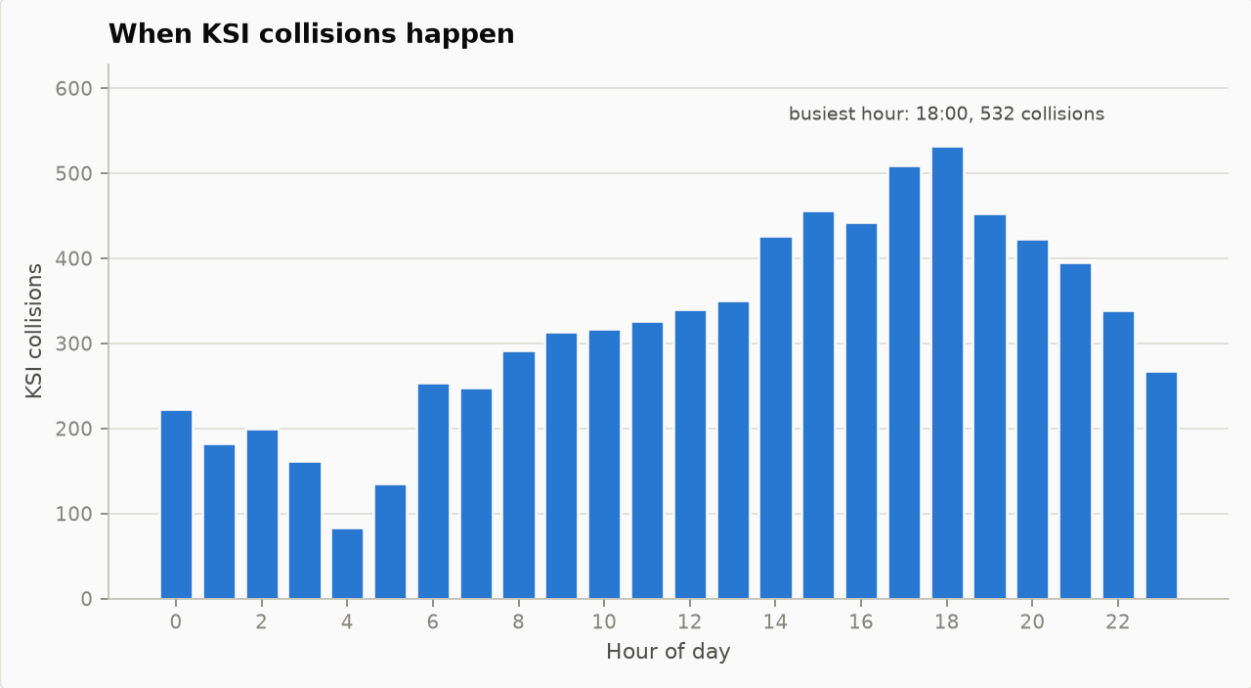
Fatality rate by road user.

A pedestrian involved in a KSI collision dies 8 times as often as a driver involved in one (16.5% against 2.1%). Pedestrians are 606 of the 1,109 people killed in the whole extract — 55% of deaths — while making up 18% of the people involved.

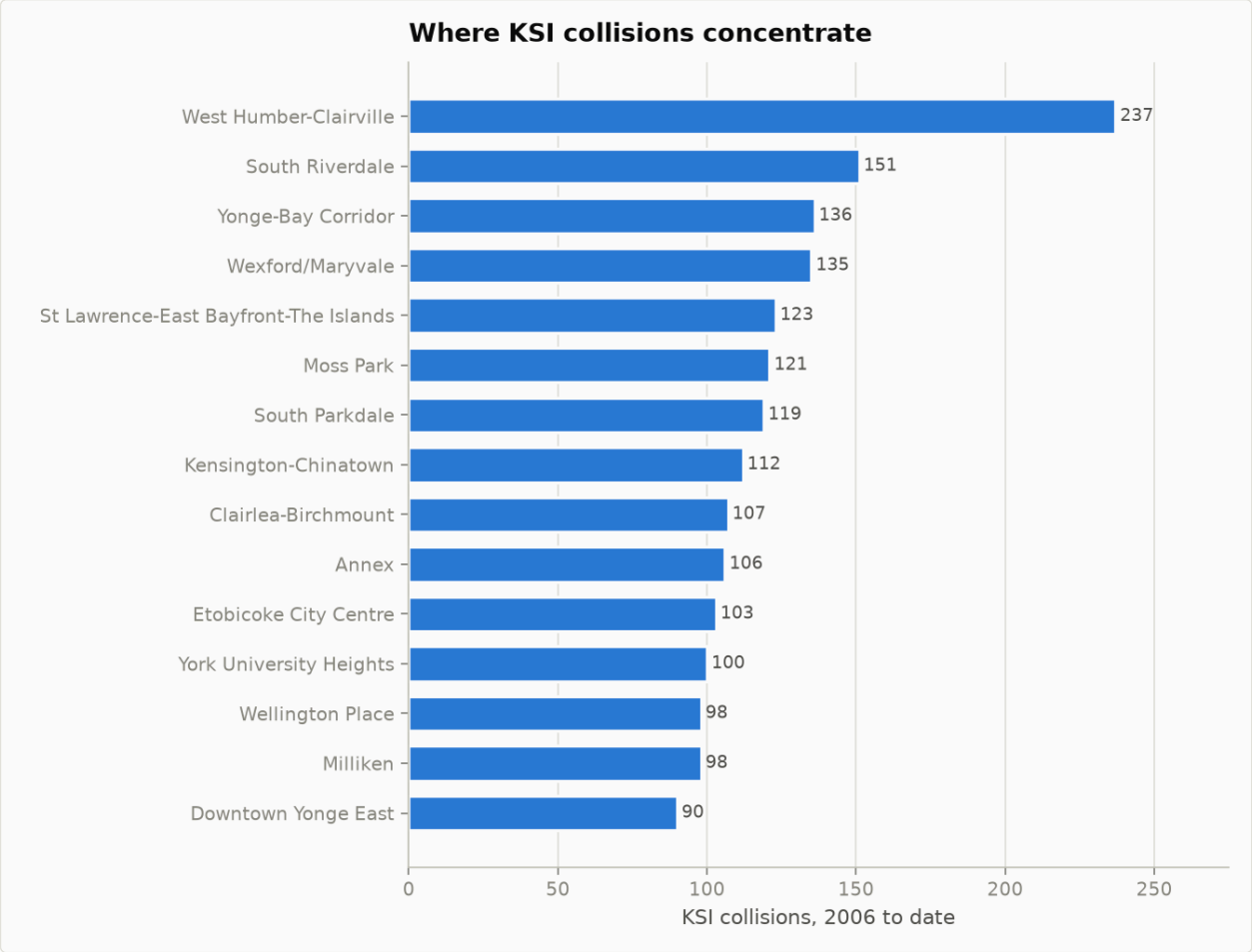
road_user	people_involved	killed	seriously_injured	fatality_rate
pedestrian	3682	606	2865	0.1646
motorcyclist	936	122	743	0.1303
cyclist	950	55	849	0.0579
passenger	3207	117	864	0.0365
driver	9977	205	1828	0.0205
owner	1828	0	0	0.0000
other	278	0	2	0.0000

The `owner` row is a data artefact rather than a finding: it records the registered owner of a vehicle, who is frequently not present, which is why the category shows zero injuries of any kind. It is left in the table so the totals reconcile.

5. When and where



KSI collisions by hour of day.



The 15 neighbourhoods with the most KSI collisions.

The neighbourhood ranking is deliberately reported as counts and not as a risk rate. A rate needs a denominator this file does not contain — resident population, or better, vehicle and pedestrian kilometres travelled through the area. Dividing by neither produces a ranking that largely reflects how busy a place is, which is worth knowing but is not the same as how dangerous it is. Treating this list as a danger ranking would be the second-biggest error available in this dataset, after the one in section 1.

neighbourhood	ksi_collisions	fatal_collisions	fatal_share
West Humber-Clairville	237	37	0.1561
South Riverdale	151	14	0.0927
Yonge-Bay Corridor	136	9	0.0662
Wexford/Maryvale	135	29	0.2148
St Lawrence-East Bayfront-The Islands	123	15	0.1220
Moss Park	121	14	0.1157
South Parkdale	119	27	0.2269
Kensington-Chinatown	112	14	0.1250
Clairlea-Birchmount	107	21	0.1963
Annex	106	9	0.0849
Etobicoke City Centre	103	17	0.1650
York University Heights	100	16	0.1600
Milliken	98	16	0.1633
Wellington Place	98	5	0.0510
Downtown Yonge East	90	7	0.0778

6. Circumstances

Each flag is a collision-level attribute, so these are collision counts. The final column compares the fatal share of flagged collisions against the fatal share of all KSI collisions.

flag	collisions	share_of_collisions	fatal_share	vs_baseline
pedestrian	3422	0.4460	0.1745	1.25
aggressive	3189	0.4156	0.1232	0.88
distracted	2248	0.2930	0.1174	0.84
older_adult	2188	0.2852	0.2020	1.44
school_child	1102	0.1436	0.1207	0.86
heavy_truck	939	0.1224	0.2396	1.71
cyclist	935	0.1219	0.0578	0.41
motorcyclist	874	0.1139	0.1384	0.99
red_light	476	0.0620	0.1218	0.87

These are associations in a file that only contains collisions serious enough to be recorded. Every rate here is conditional on a KSI collision having already happened, so none of them is the risk of anything. A flag with a high fatal share tells you which collisions turn out worst once they occur; it does not tell you what causes them or how often they occur.

7. What this analysis cannot tell you

- There is no exposure denominator. Without population, traffic volume or distance travelled, no count here can become a risk rate, and none is presented as one.
- The file contains only collisions where someone was killed or seriously injured. It cannot say anything about collisions in general, or about how often a given circumstance leads to a collision.
- Serious injury is a reported judgement and its threshold may have shifted over twenty years. This directly affects the section 3 trend.
- The circumstance flags are assigned by the reporting officer. They record what was recorded, not what happened.
- The extract is refreshed daily and figures move as historical records are amended. Everything here was computed from the snapshot identified in section 8.
- Association is not cause. Nothing in this report supports a causal claim, and none is made.

8. Reproducing this

The extract is refreshed daily, so the exact copy behind these numbers is recorded rather than described. Rerun `python run.py` against the current file and compare the snapshot to see precisely what changed.

field	value
package	motor-vehicle-collisions-involving-killed-or-seriously-injured-persons
retrieved_utc	2026-11-02T15:51:40Z
sha256	87feb1dfa7adc46170cca6ed801fa9ec8fe7ede7e62610bfd5c21541388e8761
person_rows	20882
collisions	7673
columns	50
earliest_collision	2006-01-01
latest_collision	2026-10-22

Source: City of Toronto Open Data Portal, Motor Vehicle Collisions Involving Killed or Seriously Injured Persons. Retrieved over the public CKAN API with no key or account. The data is published by Toronto Police Service and

covers the City of Toronto only.